L08 – Decision Trees

Data Mining
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Thanks to Dr Alejandro Arbelaez
And to Flaticon.com

Decision Trees

Decision Trees

- Decision models widely used for classification.

- Perfect tool to decide if you want to go out on a Saturday night!
  - Your friends call you. You don’t know if to go out or not!
  - What are the features?

Decision Trees

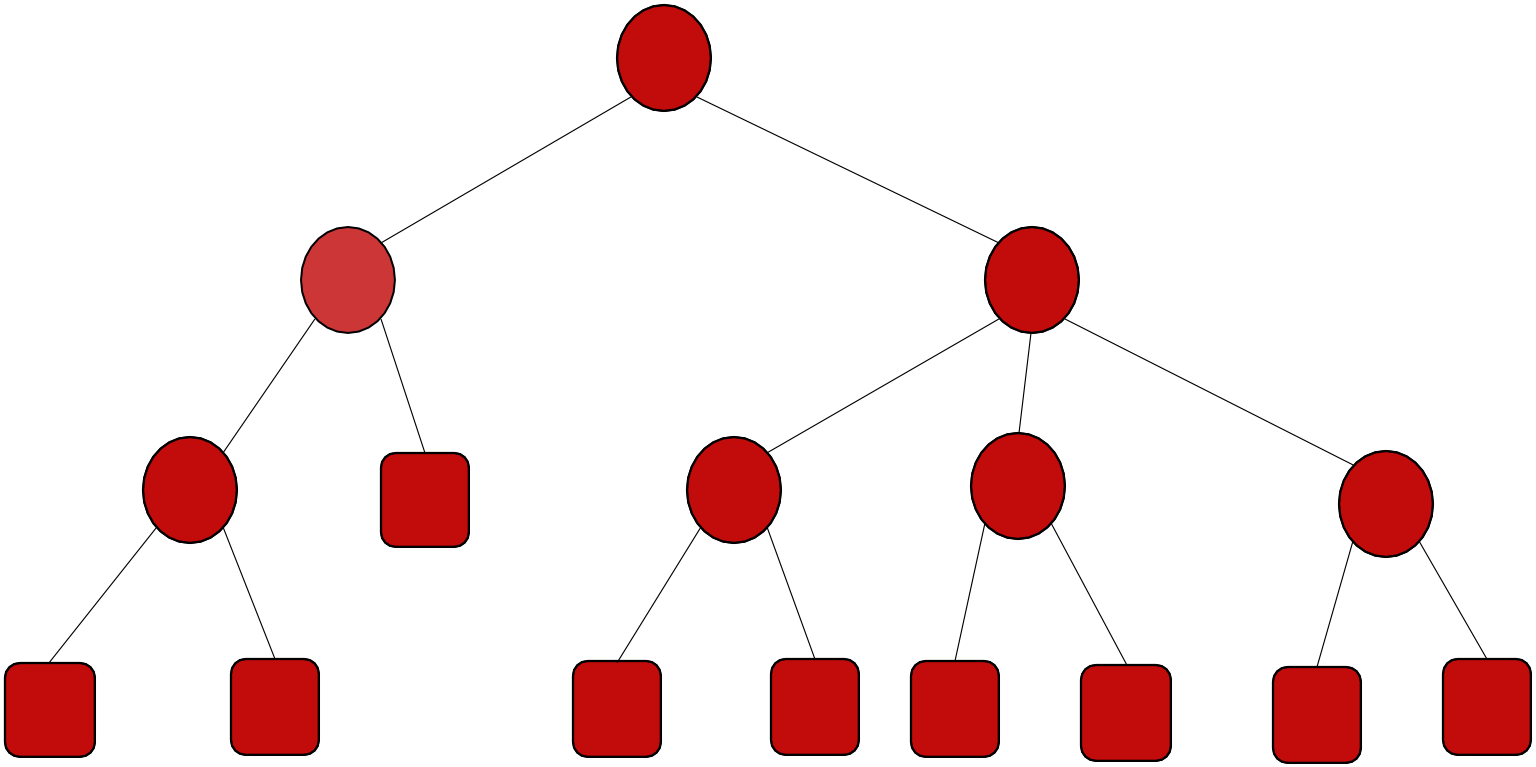
- Are you in the mood?
- How many friends are coming?
- Is it raining?
- Where are they going?
- Is the person you like going out?

What tipe of attribute are they?

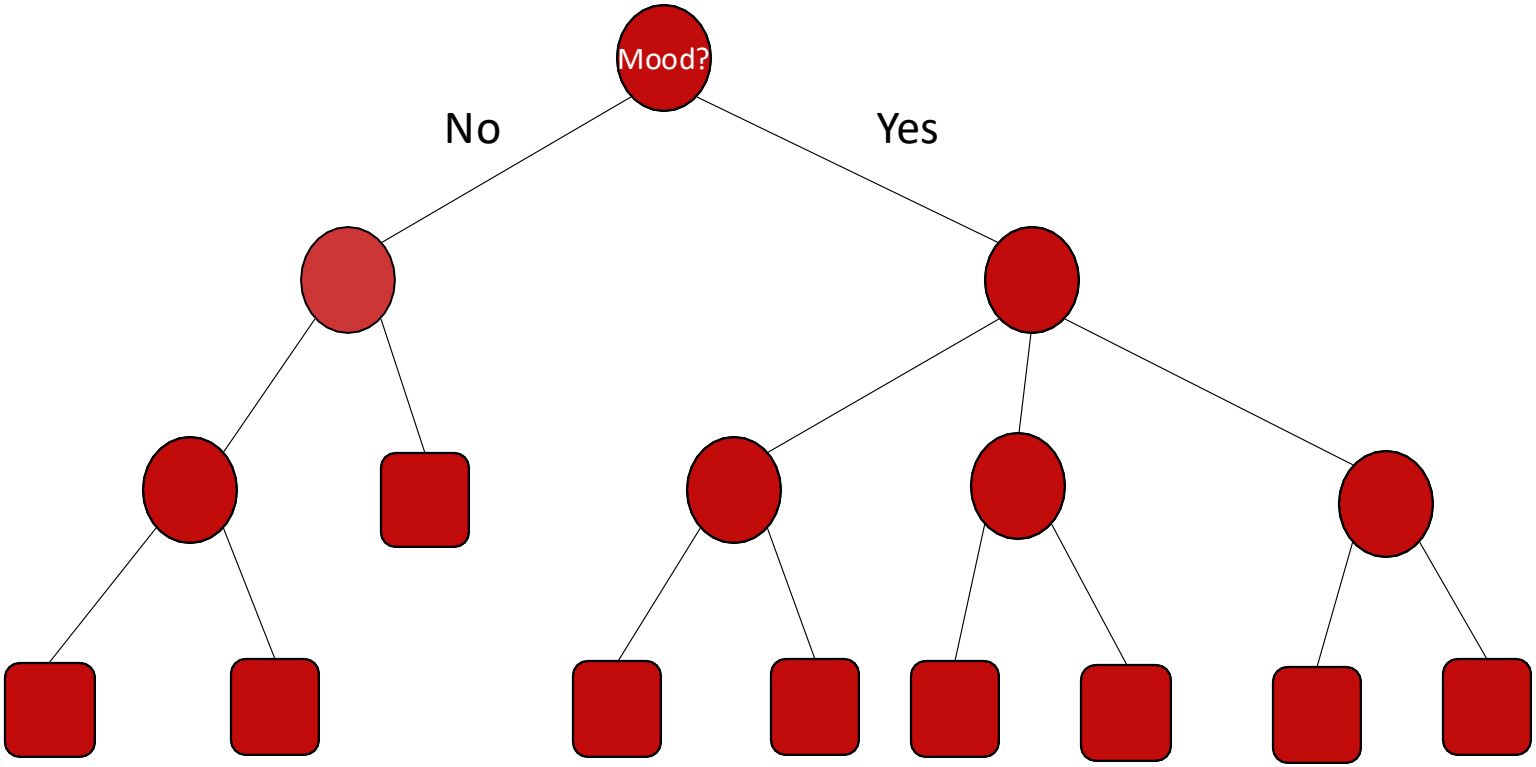
Decision Trees

Good mood	Place	Rainy	Person you like	Number of Friends	Decision
Yes	Voodoo	Yes	No	7	?
No	Old Oak	Yes	No	4	?
Yes	Old Oak	No	No	5	?
No	An Brog	No	No	9	?
Yes	Voodoo	Yes	Yes	2	?

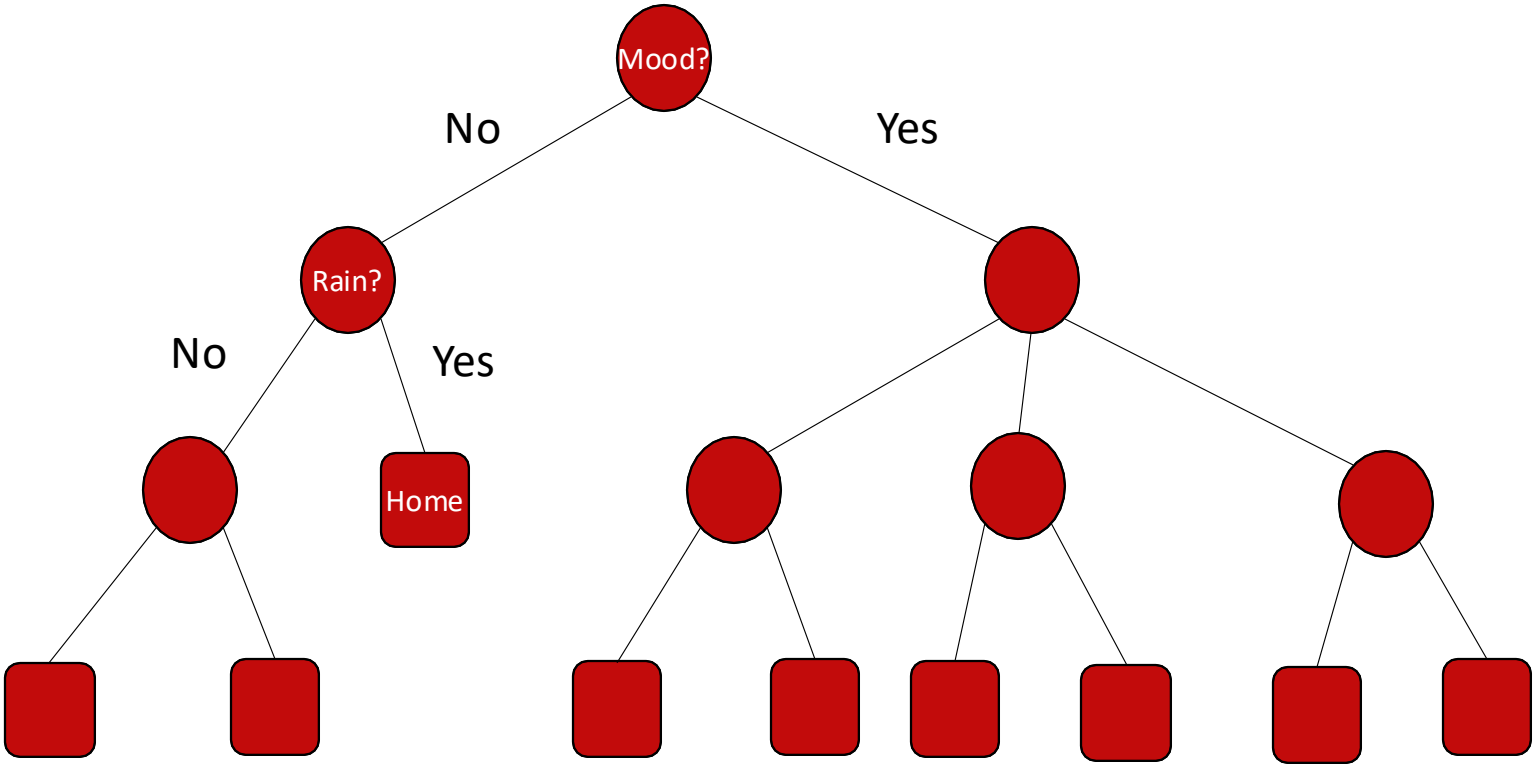
Decision Trees



Decision Trees



Decision Trees



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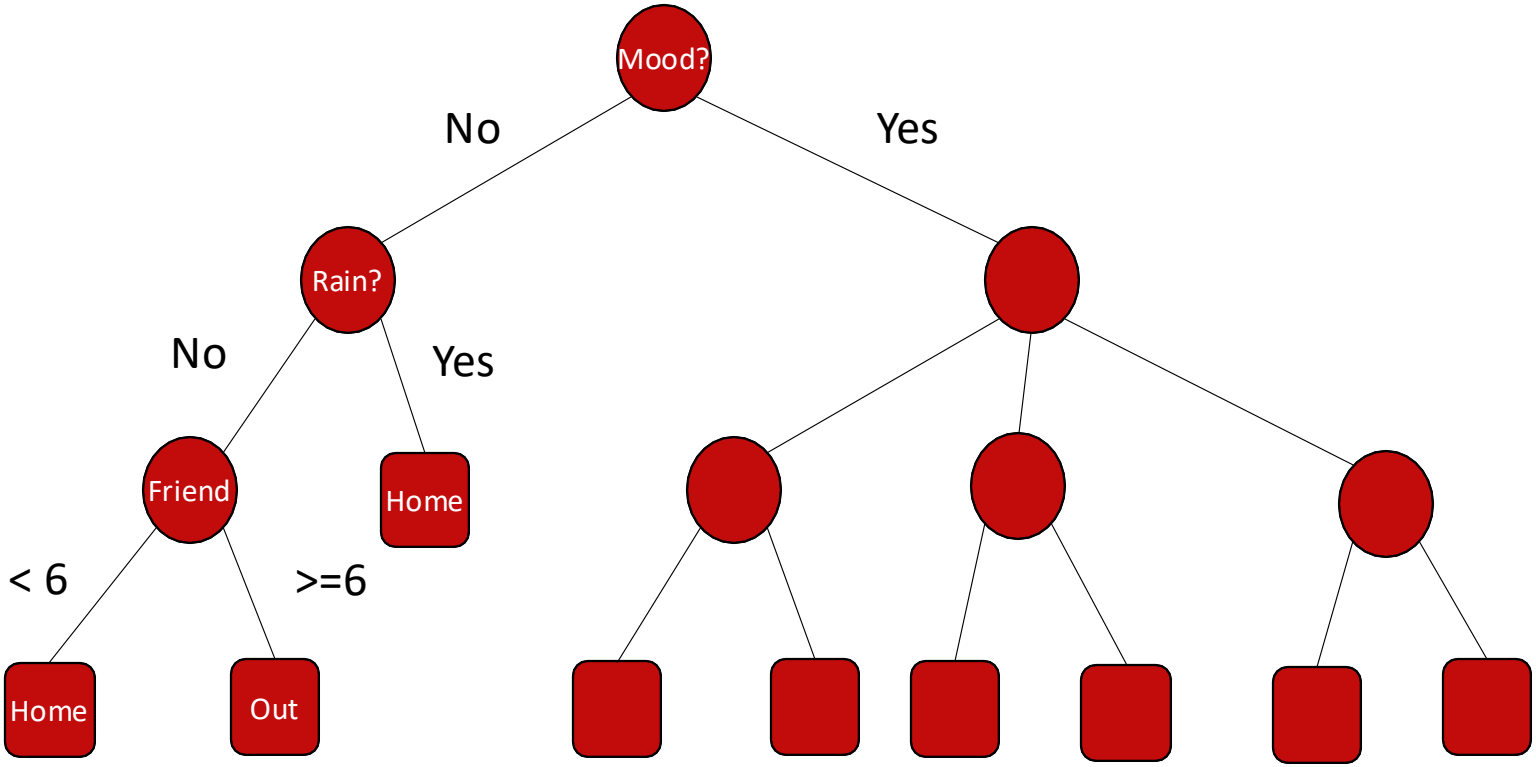

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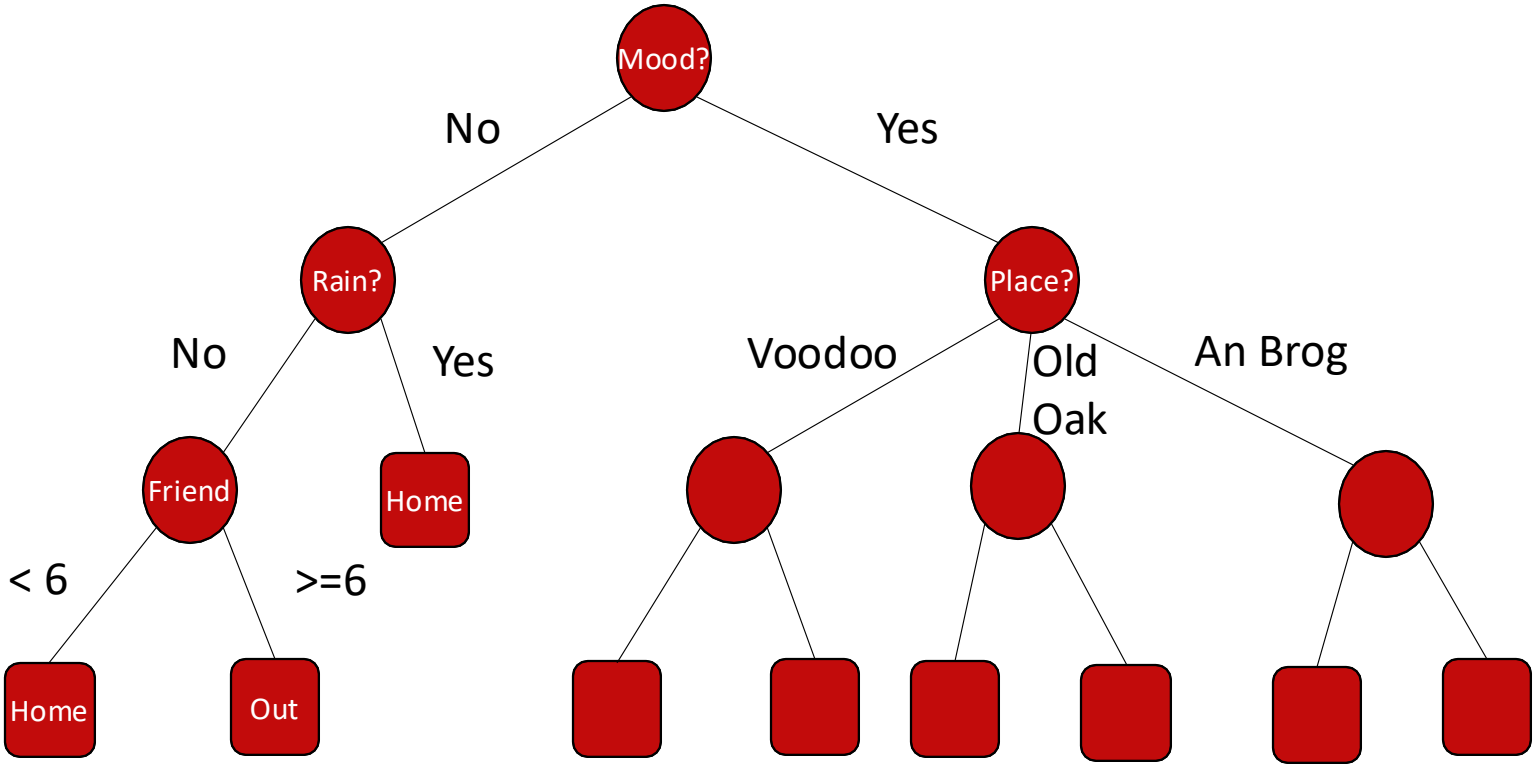
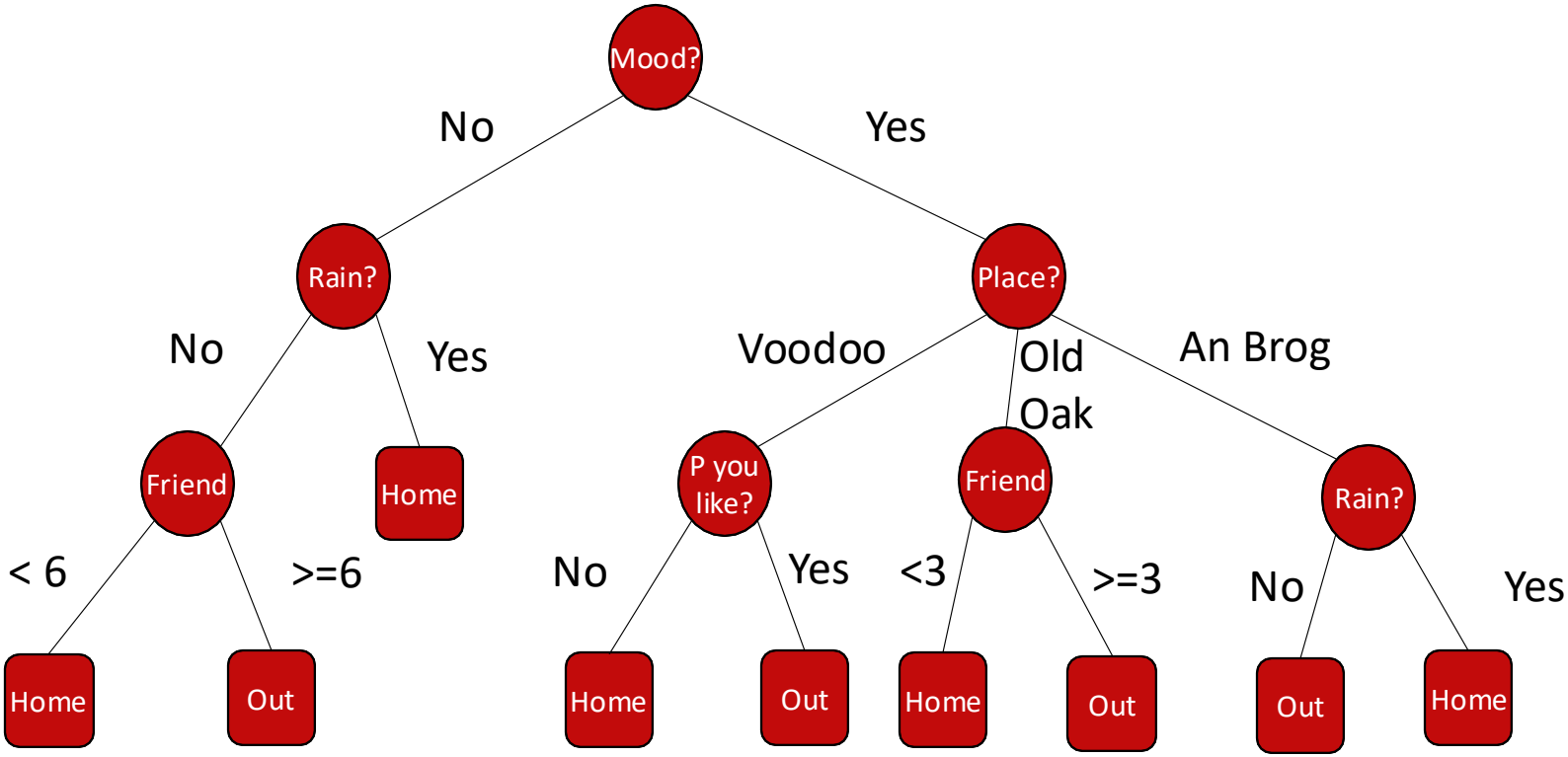
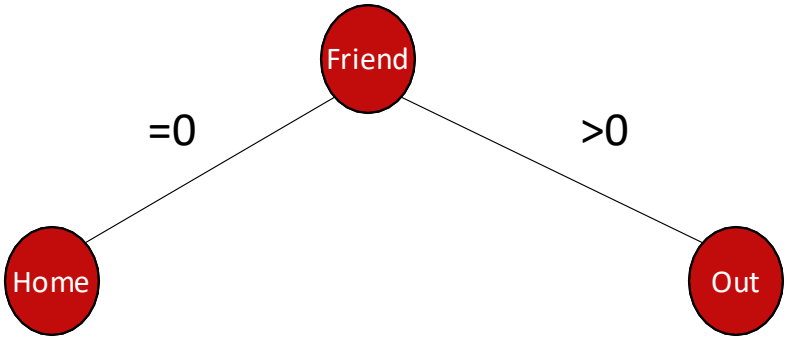
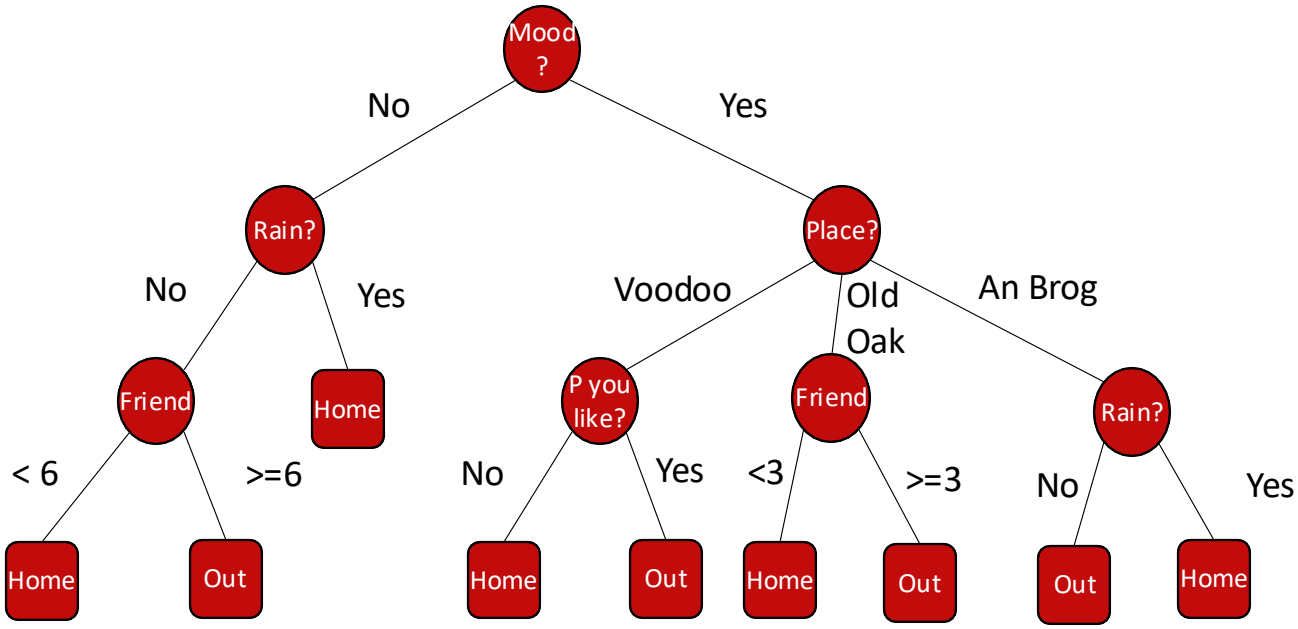


Decision Trees



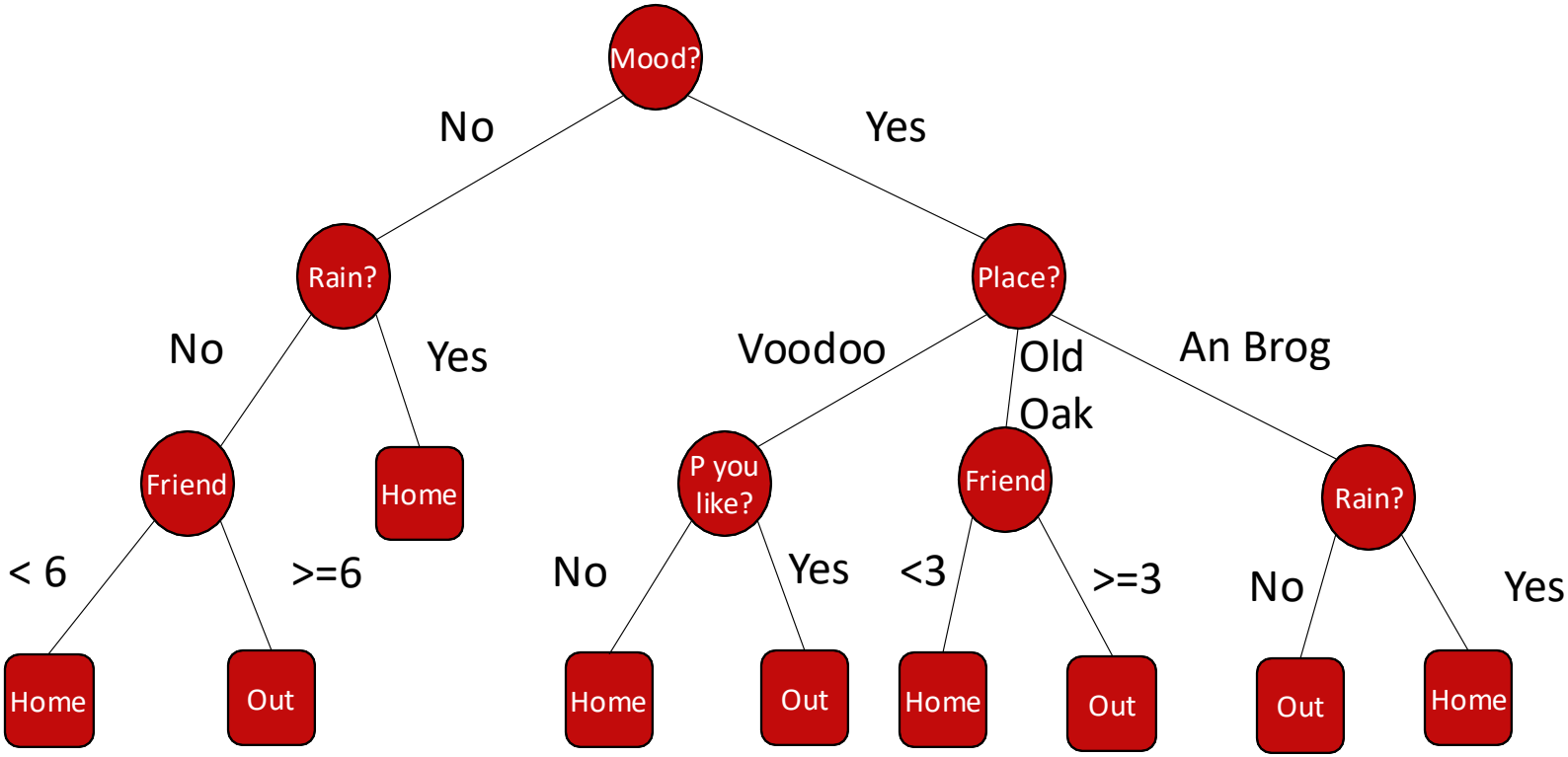
Decision Trees





Good mood	Place	Rainy	Person you like	Number of Friends	Decision
Yes	Voodoo	Yes	No	7	?
No	Old Oak	Yes	No	4	?
Yes	Old Oak	No	No	5	?
No	An Brog	No	No	9	?
Yes	Voodoo	Yes	Yes	2	?

This is the ML model!



Decision Trees

- What could the training set be?

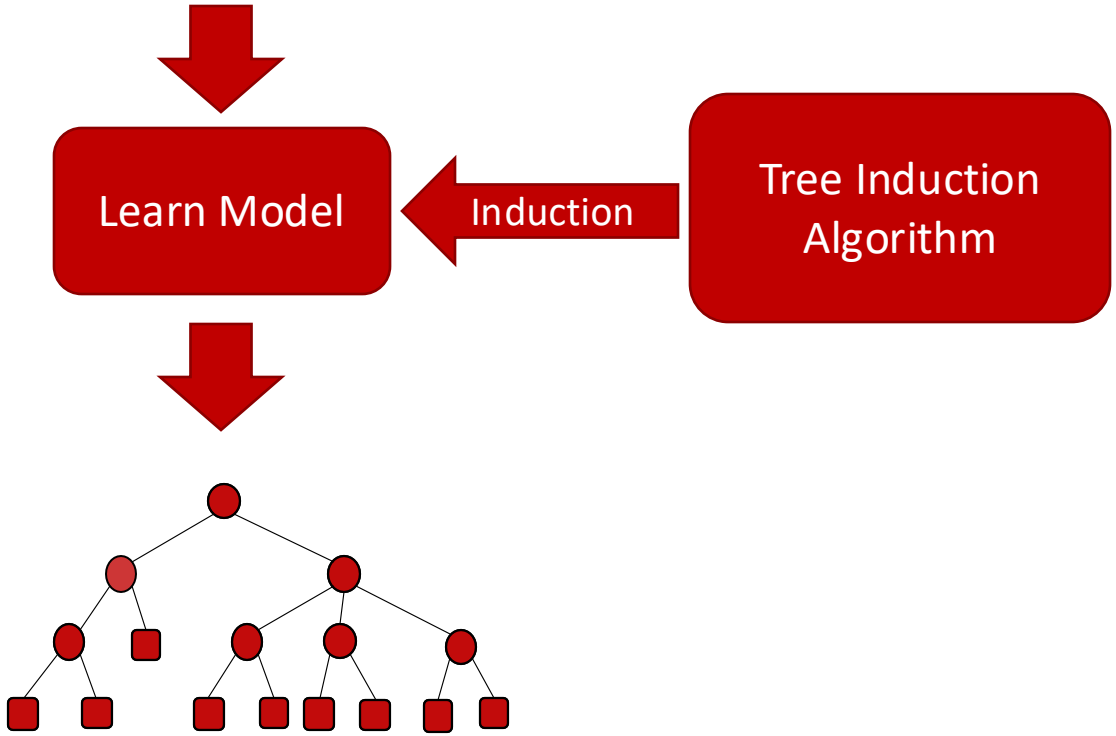
Decision Trees

- What could the training set be?

Good mood	Place	Rainy	Person you like	Number of Friends	Did I have fun?
Yes	Voodoo	No	No	7	Yes
No	An Brog	Yes	No	10	No
No	Voodoo	No	Yes	9	Yes
Yes	Old Oak	No	No	1	No
Yes	Voodoo	No	No	2	No

Building this tree

Good mood	Place	Rainy	Person you like	Number of Friends	Fun?
Yes	Voodoo	No	No	7	Yes
No	An Brog	Yes	No	10	No
No	Voodoo	No	Yes	9	Yes
Yes	Old Oak	No	No	1	No
Yes	Voodoo	No	No	2	No

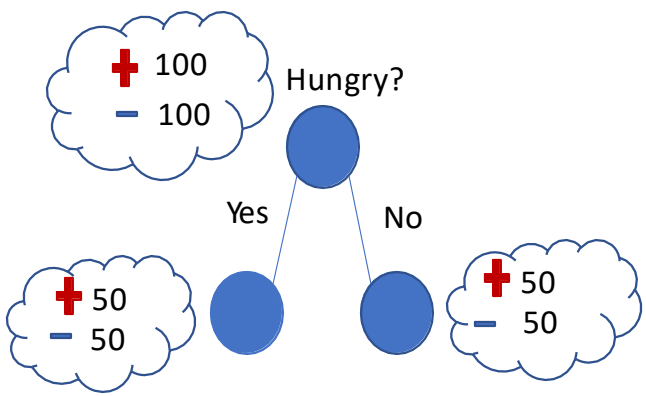


Decision Tree Learning

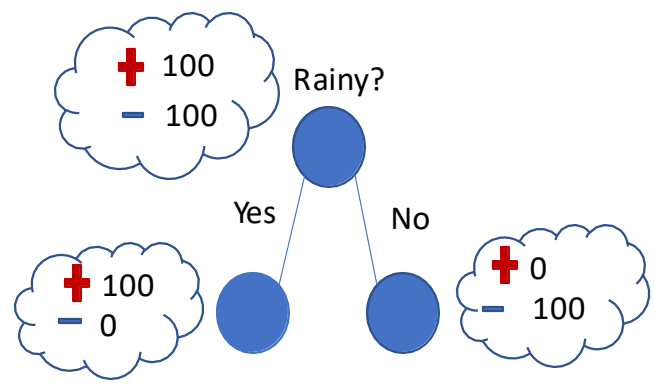
1. Pick the best attribute
   - Best: Split the data
2. Ask question
3. Follow the path
4. Go to 1 until you got an answer

Decision Trees

- Impurity Measure
  - Which split is better?



100% impure
A node is split 50/50



100% pure
When all of its data points belongs to a single class

Entropy

- Entropy is a very important concept in understanding decision trees and is defined as the level of uncertainty in a random variable.
- For example, a coin that always comes up heads has no uncertainty and therefore has an entropy value of 0.
- Entropy is 0 if all members of the dataset belong to the same class (the data is perfectly classified).
- The range of entropy is 0 ("perfectly classified") to 1 ("totally random").

Entropy

https://www.youtube.com/watch?v=MYdBJG6Lbfc

Entropy

- So let's now look at the following attributes and identify how well they reduce the overall entropy.

Rain	Swell	Surf
Yes	Big	Yes
Yes	Small	No
No	Small	No
No	Big	Yes

Entropy

Entropy 0 ("perfectly classified")
Entropy 1 ("totally random")

- You can think of entropy of how well classified a subset of data is.
- The Swell attribute has a low entropy level in this case because there is little uncertainty.
  - Ent(Surf, Swell = Big) = Ent(2+, 0-) = 0
- Every time the swell is big we go surf.
- The swell attribute has two values. What is the entropy for Swell = Small

Rain	Swell	Surf
Yes	Big	Yes
Yes	Small	No
No	Small	No
No	Big	Yes

Entropy

Entropy 0 ("perfectly classified")
Entropy 1 ("totally random")

- Rain for example has an entropy of 1 because it has quite a high level of associated uncertainty.
  - Ent(Surf, Rain = Yes) = Ent(1+, 1-) = 1.
- In other words when it was raining, we went surfing on one occasion and once we didn't go surfing. Therefore the influence of this variable on whether or not we go surfing is highly uncertain.
- What is the entropy for the rain attribute when it's value is No.

Rain	Swell	Surf
Yes	Big	Yes
Yes	Small	No
No	Small	No
No	Big	Yes

Entropy

Entropy 0 ("perfectly classified")
Entropy 1 ("totally random")

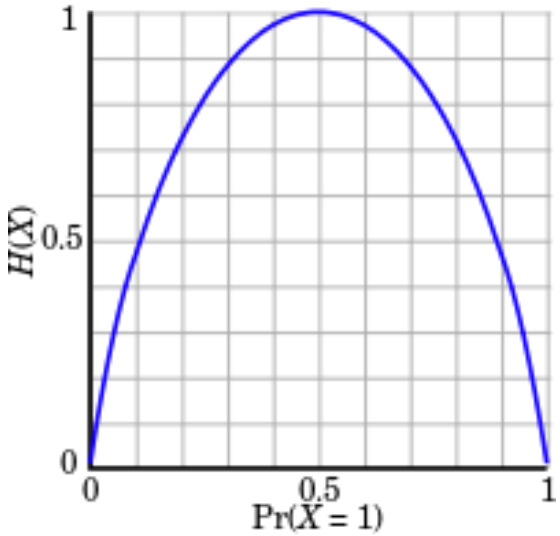
We can calculate the entropy of the dataset S as:

$$Ent(S) = H(S) = \sum_{i=1}^{n} -p_i \log_2 p_i$$

- For a dataset S with n different classes(outputs), proportion of class i is p_i (Number of instance of class i divided by total number of instances)

Note we can use the same formula for calculating the entropy for various attribute values such as Ent(Swell = Big)

Entropy



$$\mathrm{Ent}(S) = \sum_{i=1}^{n} - p_i \log_2 p_i$$

Entropy Example

Entropy 0 ("perfectly classified")
Entropy 1 ("totally random")

In the example below we will determine the entropy for the entire dataset S

$$Ent(S) = H(S) = \sum_{i=1}^{n} -p_i \log_2 p_i$$

- There are two attributes in our dataset, therefore n=2.

Ent(Surf) = ((-2/4) log2(2/4)) + ((-2/4) log2(2/4))= 1 (High Level of Uncertainty)

Rain	Swell	Surf
Yes	Big	Yes
Yes	Small	No
No	Small	No
No	Big	Yes

Entropy Example

Entropy 0 ("perfectly classified")
Entropy 1 ("totally random")

- Now lets examine the entropy of the swell attribute when Swell = Big. n
- Ent(Swell=Big) = (-2/2) log2(2/2) = 0 (Lowest uncertainty, no uncertainty in this data)

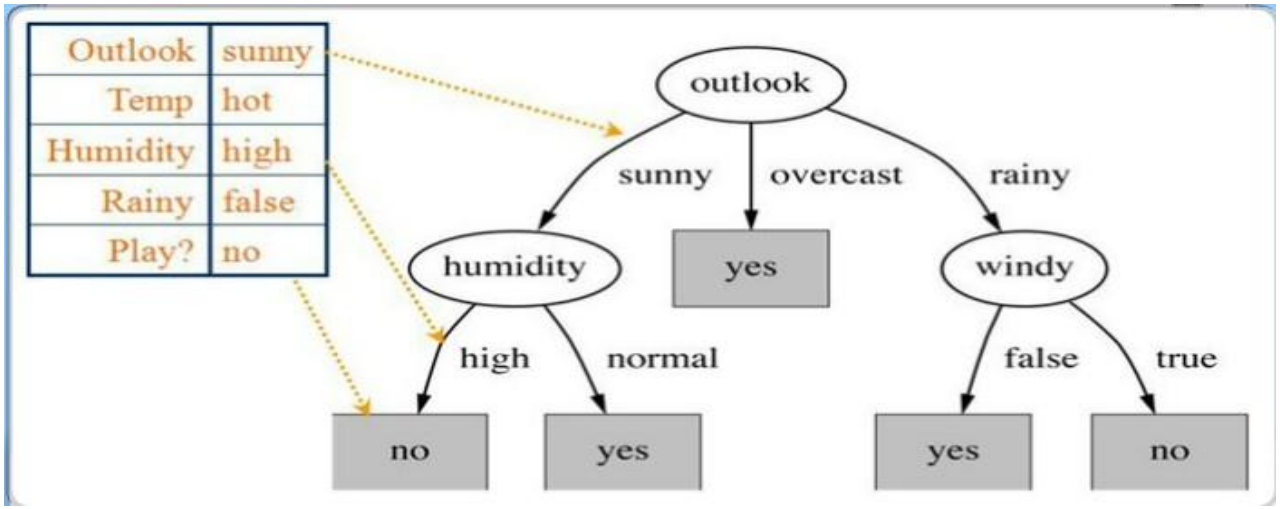
Rain	Swell	Surf
Yes	Big	Yes
Yes	Small	No
No	Small	No
No	Big	Yes

Weather Dataset

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Weather Dataset

- Each node of the tree is an attribute test. The value of the attribute determines which branch to follow. We continue until we reach a leaf which has a decision.



Ross Quinlan – ID3 Algorithm

- Ross Quinlan is a computer science researcher in data mining and decision theory

- Provided significant contributions to the area of decision tree algorithms

- Invented the C4.5 and ID3 algorithms for inducing decision trees

Overview of ID3

Step 1

For all attributes that have not yet been used in the tree calculate their **entropy** and **information gain** values for the test samples

Step 2

Select the attribute that has the highest information gain

Step 3

Make a tree node containing that attribute

Entropy Examples

- In the following slides we will look more closely at the entropy calculations for our weather dataset:
  - **Example A** – We demonstrate how to calculate the **entropy value of the dataset as a whole**, in other words the entropy of the weather dataset S.
  - **Example B** – Demonstrate how to calculate the **entropy value of the Windy attribute**. We calculate the entropy for Windy = false and for Windy = true

Weather Dataset

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Entropy Example A – Entire Dataset

n

- The dataset S refers to our Weather dataset, which has two classes (yes and no).
- We must first determine the proportion of each class i (p_i) in S.
  - There is a total of 14 examples in the dataset (the class **yes** occurs 9 times and the class **no** occurs 5 times.)
  - Therefore, the portion of class **yes** is simply 9/14 and the portion of class **no** is 5/14. We can now plug these values into the formula and calculate the entropy of the weather dataset.

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Entropy Example A – Entire Dataset

$$Ent(S) = \sum_{i=1}^{n} -p_i \log_2 p_i$$

- S is the weather dataset

```
Ent(S) = Ent([9+,5-])
Ent(S) = -9/14 log2(9/14) - 5/14 log2(5/14)
Ent(S) = 0.940
```

Entropy Example B – Windy Attribute

- If we were to divide the data based on the attribute **Windy**. We could need to assess how well this attribute splits the data.
- We need to calculate the entropy for the **Windy = true** subset and the
- **Windy = false** subset.
- In order to calculate the entropy for the **Windy = true** subset we must extract the result (play tennis 'yes' or 'no') when **Windy = true**.
  - This subset is highlighted in green on the next slide.
  - In order to calculate the entropy for the Windy = false subset we must extract the result when Windy = false. This subset is highlighted in yellow in the slide.

Entropy Example B

Anyone for Tennis?					
ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
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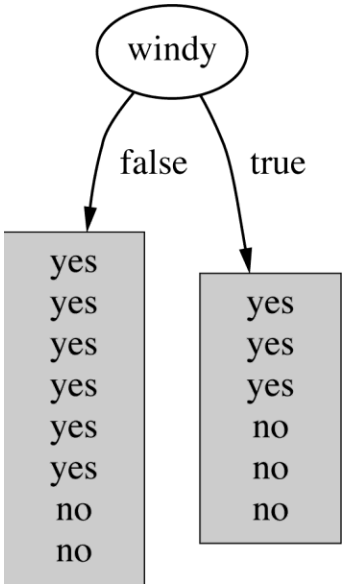
Entropy Example B

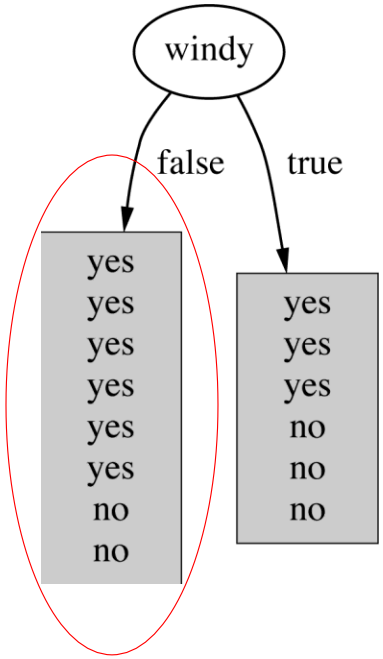


- The diagram shows the division of the Weather dataset based on the value of the attribute windy.

- The grouping on the column on the left shows the output (play tennis ‘yes’ or ‘no’) for when **Windy = false** and corresponds to the subset highlighted in green in the previous slide.

- The grouping on the right shows the output for when **Windy = true** and corresponds to the subset highlighted in yellow in the previous slide.

Entropy Example B

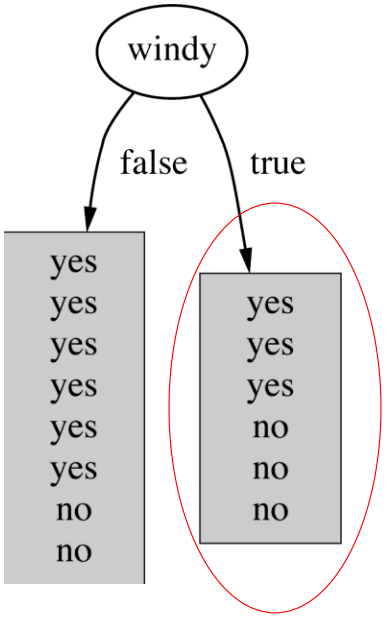


- We now have the information we need to calculate the entropy for **Windy = true** and **Windy = false**.

- The **Windy = false** subset we have a total of eight entries, 6 are positive ('yes') and 2 are negative ('no'). Therefore, we calculate the Entropy (Ent([6+, 2-]))

$$\mathrm{Ent}(S|\mathrm{Windy}=\mathrm{false}) = \mathrm{Ent}([6+,2-])$$
$$= -6/8\ \log_2(6/8) - 2/8 \log_2(2/8)$$
$$= 0.811$$

Entropy Example B

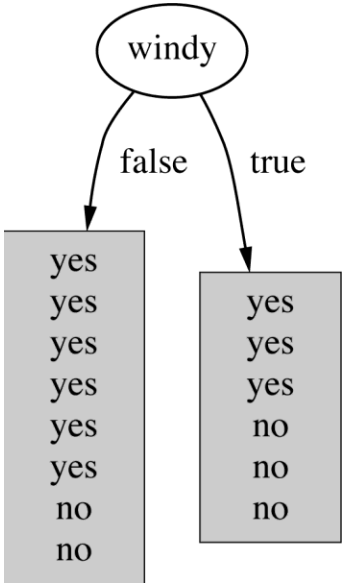


- The **Windy = true** subset we have a total of six entries, 3 are positive ('yes') and 3 are negative ('no'). Therefore, we calculate the Entropy (Ent([3+, 3-])). What would you expect the entropy value to be?

$$\mathrm{Ent}(S|\mathrm{Windy}=\mathrm{true})=\mathrm{Ent}([3+,3-])$$
$$=-3/6\log_2(6/8)-3/6\log_2(6/8)$$
$$= 1$$

Entropy Example B

Ent(S)=0.940



Ent(S|Windy=false)=0.811

Ent(S|Windy=true)=1

Weather Dataset

ID	Outlook	Temp	Humidity	Windy	Play?
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K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Entropy Example B

- Let's focus on the Humidity attribute

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
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D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
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K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Entropy Example B

- When **Humidity = normal** we have 6 yes and 1 no

- When **Humidity = high** we have 3 yes and 4 no

Entropy Example B

- When **Humidity = normal** we have 6 yes and 1 no

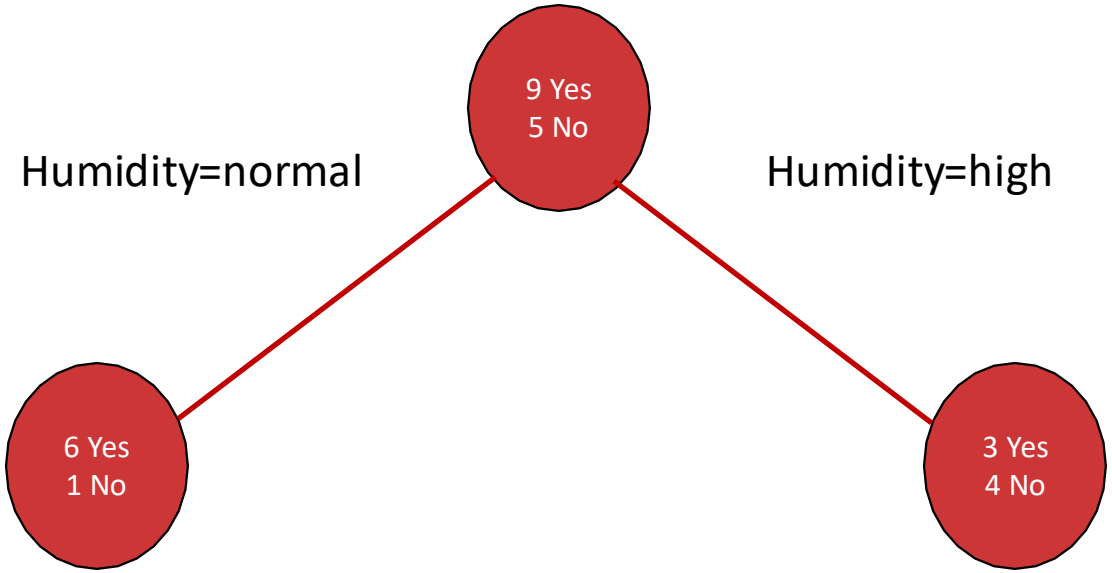
```
Ent(S|Humidity=normal) = Ent([6+,1-])
    = 0.591
```

- When **Humidity = high** we have 3 yes and 4 no

```
Ent(S|Humidity=high) = Ent([3+,4-])
    = 0.985
```

Entropy Example B

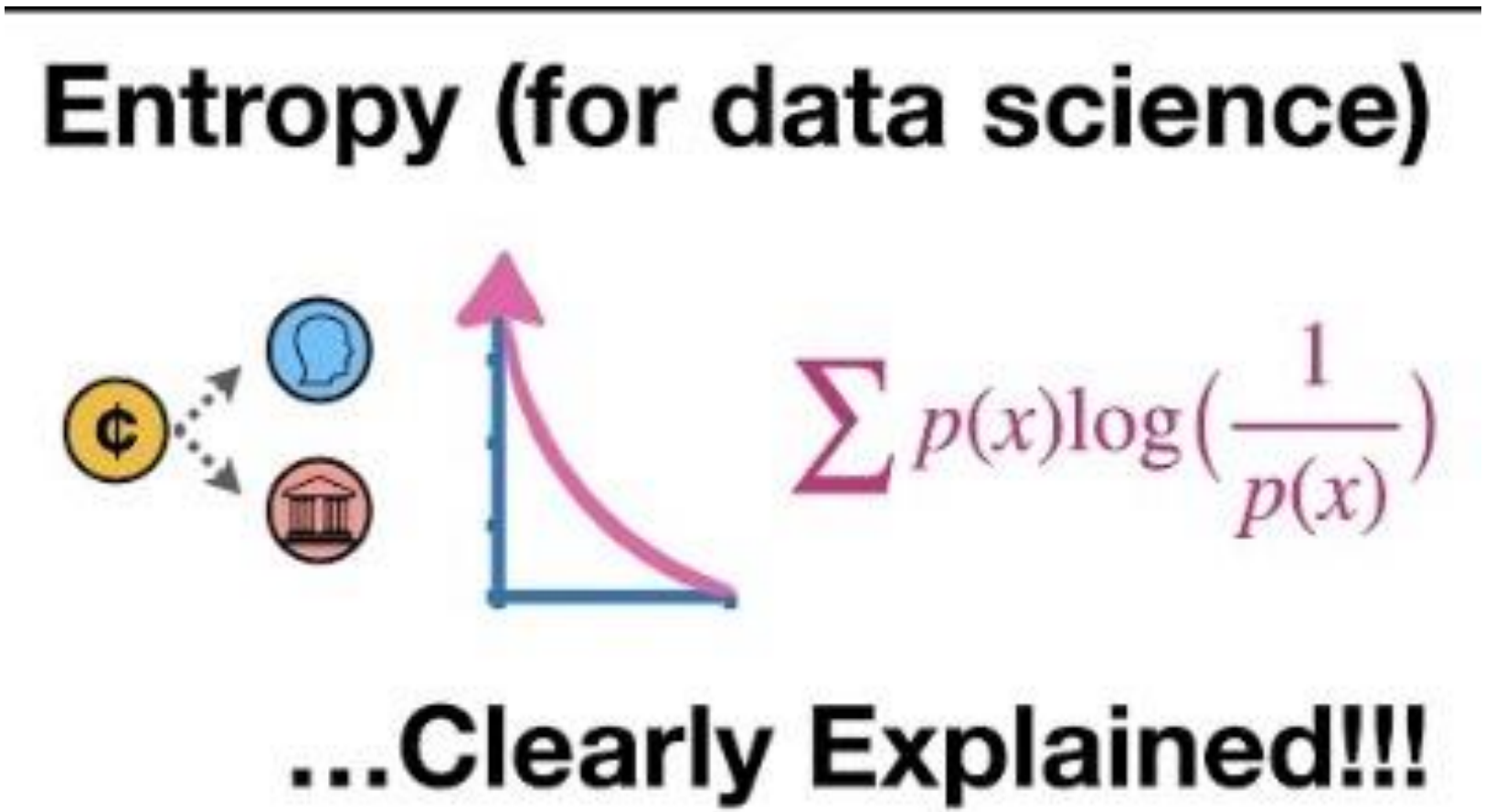
Ent(S)=0.940



Ent(S|Humidity=normal)=0.591

Ent(S|Humidity=high)=0.985

Interesting video on Entropy



Overview of ID3

Step 1

For all attributes that have not yet been used in the tree calculate their **entropy** and **information gain** values for the test samples

Step 2

Select the attribute that has the highest information gain

Step 3

Make a tree node containing that attribute

Information Gain

- Information gain of a attribute is the reduction in entropy from dividing the data into two or more subsets based on that attribute.

Information Gain

- **Information gain** of a attribute is the **reduction in entropy** from dividing the data into two or more subsets based on that attribute.
- Therefore, in our Weather example the information gain of the attribute Temperature would be the overall reduction in entropy caused by partitioning the data into three subsets:
  1. Those where the value is hot;
  2. Those where the value is mild;
  3. Those where the value is cool.

Information Gain

- Ultimately, we are looking for the attribute that would lead to the **highest information gain** (or put another way that would give the **largest reduction in entropy**).
  - The higher the information gain of a particular attribute then the better that attribute partitions the data.

Information Gain

- Information Gain of an attribute is the reduction in Entropy from partitioning the data according to that attribute.

$$Gain\ (S, A) = Ent(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Ent(S_v)$$

Information Gain

Cardinality: number of elements of S where A=v

Gain for attribute A of set S

$$Gain\ (S, A) = Ent(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Ent(S_v)$$

Number of elements in S

For all the possible values v in A

Information Gain

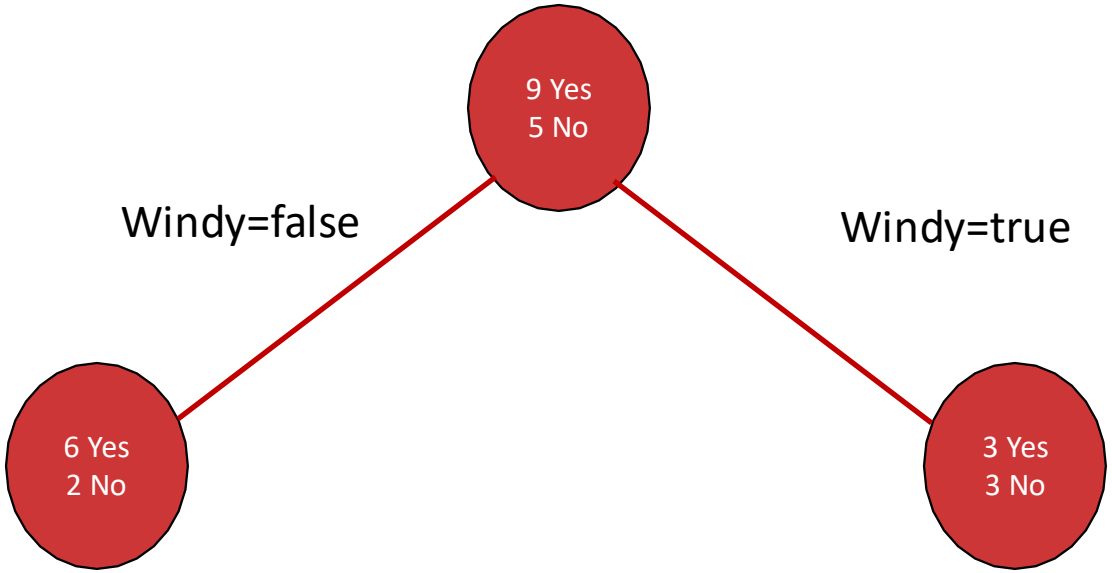
$$Gain\ (S, A) = Ent(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Ent(S_v)$$

- Size of S

- Lets take the attribute *A* as being Windy
- The entropy of each value *v* of windy is calculated (Windy = true, Windy = false)
- They are then added together **proportionately according to how big of a fraction of the overall dataset they each are** (this is what the ratio $\frac{|S_v|}{|S|}$ is for in the formula)

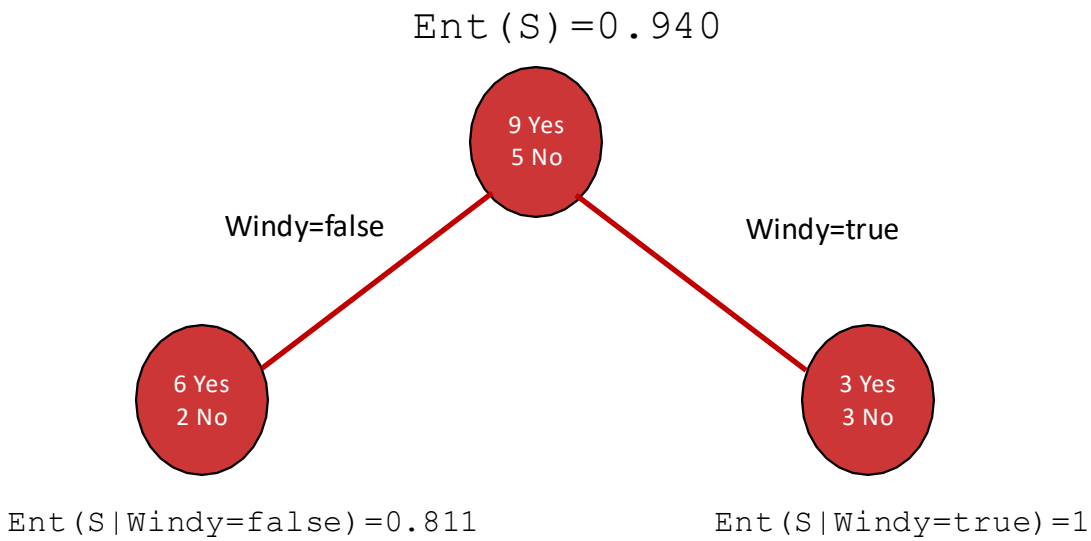
Entropy Example B

Ent(S)=0.940



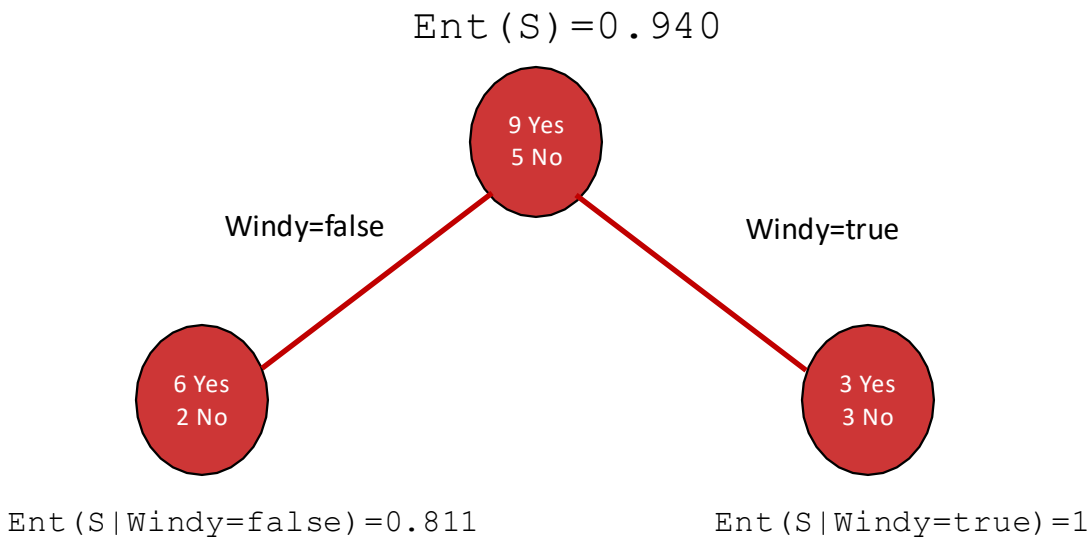
Ent(S|Windy=false)=0.811

Ent(S|Windy=true)=1



$$Gain\ (S, A) = Ent(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Ent(S_v)$$

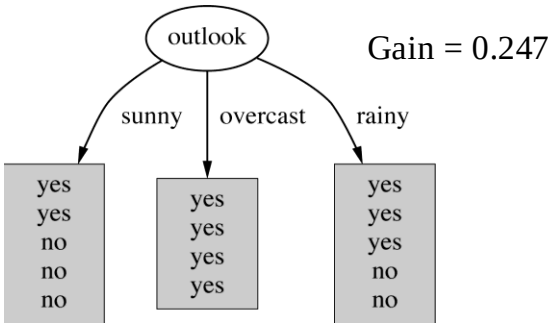
$$Gain\ (S, Windy) = Ent(S) - \sum_{v \in \{true, false\}} \frac{|S_v|}{|S|} Ent(S_v)$$



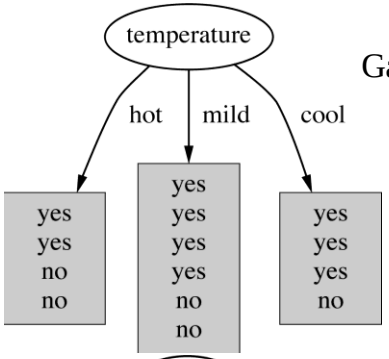
$$Gain\ (S, Windy) = Ent(S) - \sum_{v \in \{true, false\}} \frac{|S_v|}{|S|} Ent(S_v)$$

$$Gain\ (S, Windy) = 0.940 - \frac{8}{14} 0.811 - \frac{6}{14} 1.00$$
$$= 0.048$$

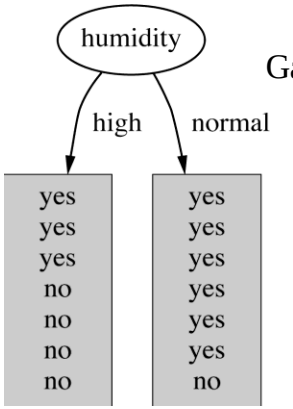
Information Gain Example



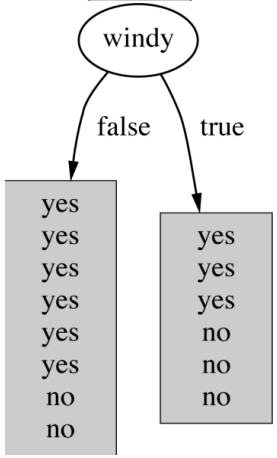
Gain = 0.247



Gain = 0.029



Gain = 0.152



Gain = 0.048

Overview of ID3

Step 1

For all attributes that have not yet been used in the tree calculate their **entropy** and **information gain** values for the test samples

Step 2

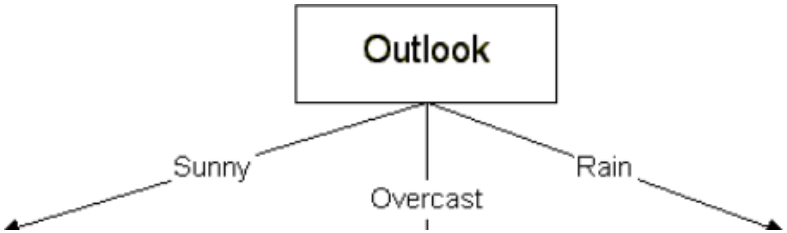
Select the attribute that has the highest information gain

Step 3

Make a tree node containing that attribute

Adding Nodes to the Decision Tree

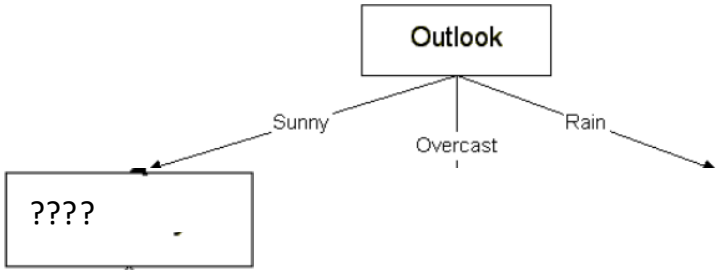
- The previous slides shows the information gain for each attribute in the weather dataset.
- The next step is to "**select the attribute that has the highest information gain**" and make it the root node of our decision tree.
- The outlook attribute has the highest information gain value and therefore becomes a root node.
- We split our tree based on the three possible values of the outlook attribute



Adding Nodes to the Decision Tree

- Lets find the best attribute to place on the Sunny branch of our decision tree below.
- To do this we will need to calculate the information gain for all remaining attributes (Humidity, Temperature, Windy) for the data where Outlook = Sunny

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no



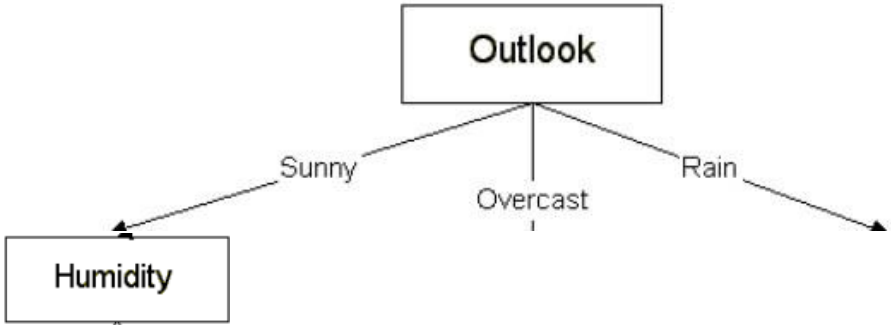
Information Gain for Humidity Attribute

- Ent(Sunny) = $-((2/5) * \log_2(2/5)) - ((3/5) * \log_2(3/5))$
  = 0.97095059445
  - (5 Data items where Sunny = yes and no (3 where it is no and 2 where it is yes))
- Ent(Sunny, Humidity = High) = Ent(0+, 3-) = 0
  - (3 data items where sunny and humidity = high, of those 3 they all had an outcome of play = no)
- Ent(Sunny, Humidity = Normal) = Ent(2+, 0-) = 0
  - (2 data items where sunny and humidity = normal, of those 2 they all have an outcome of play = yes)
- Gain(Sunny, Humidity) = 0.97095059445 - ((3/5)0 + (2/5)0) = 0.97095059445

ID	Outlook	Humidity	Play?
A	sunny	high	no
B	sunny	high	no
C	overcast	high	yes
D	rainy	high	yes
E	rainy	normal	yes
F	rainy	normal	no
G	overcast	normal	yes
H	sunny	high	no
I	sunny	normal	yes
J	rainy	normal	yes
K	sunny	normal	yes
L	overcast	high	yes
M	overcast	normal	yes
N	rainy	high	no

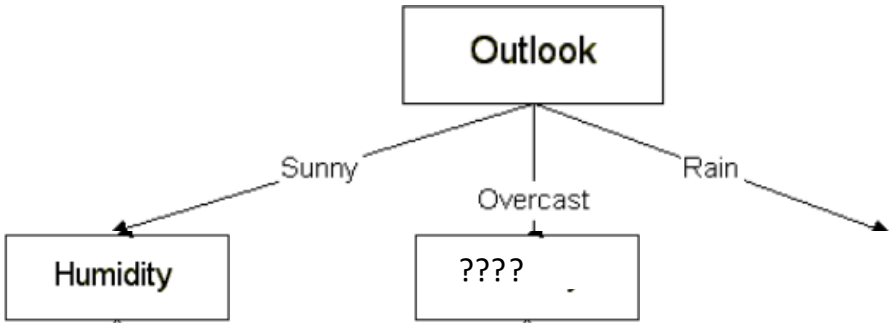
Adding Nodes to the Decision Tree

- Lets now look at applying ID3 to the subset of the dataset where Outlook is Sunny
  - There are 5 examples from table 1 with outlook = sunny
  - Gain(Sunny, Humidity) = 0.970
  - Gain(Sunny, Temperature) = 0.570
  - Gain(Sunny, Wind) = 0.019
- What should be the new node added to the decision tree



Adding Nodes to the Decision Tree

Lets find the best attribute to place on the **Overcast** branch of our decision tree below.



Adding Nodes to the Decision Tree

- Any observations on the subset of the dataset where outlook is overcast?

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Adding Nodes to the Decision Tree

- When we come to apply ID3 to the subset where Outlook=overcast we will find that further partitioning of this subset is not needed, as all the cases have the same class

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
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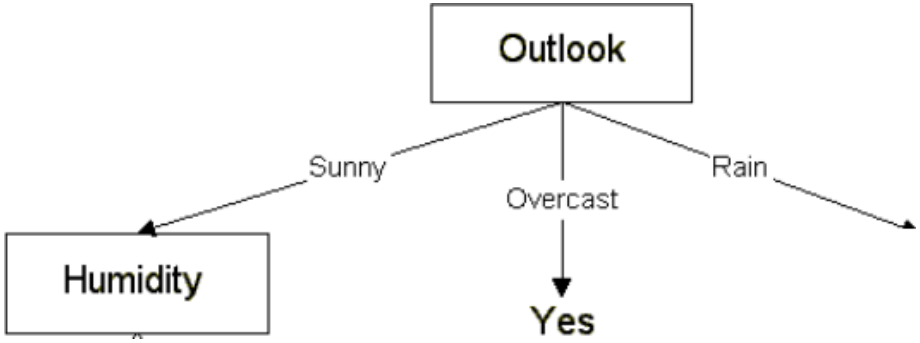
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Adding Nodes to the Decision Tree



Adding Nodes to the Decision Tree

- Any observations on the subset of the dataset where outlook is overcast?

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M	overcast	hot	normal	false	yes
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Adding Nodes to the Decision Tree

- Lets find the best attribute to place on the Sunny branch of our decision tree below.
- To do this we will need to calculate the information gain for all remaining attributes (Humidity, Temperature, Windy) for the data where Outlook = Sunny

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I	sunny	cool	normal	false	yes
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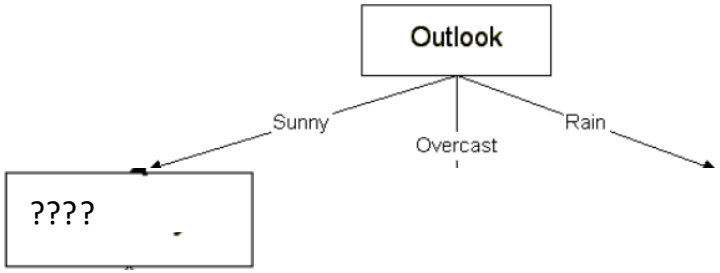
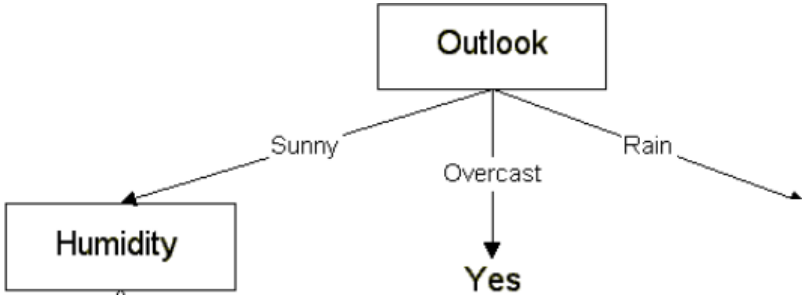
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no



Exercise

- Determine the structure of our decision tree after the humidity node.



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Exercise

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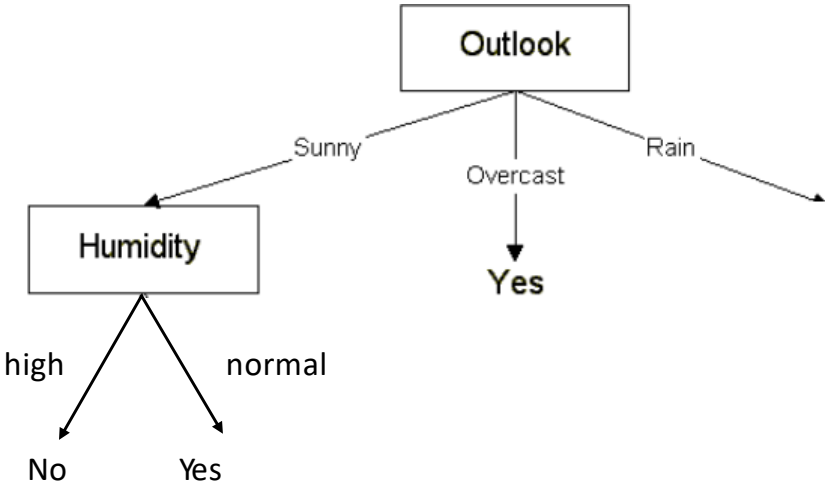

ID	Outlook	Temp	Humidity	Windy	Play?
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K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

ID3 Algorithm

```
ID3(Examples, Attributes, Target):
Input:        Examples: set of classified examples
       Attributes: set of attributes in the examples
       Target:       classification to be predicted
if all Examples have same class then return this class

else if all Attributes are tested then return majority class

else:
       let Best = chooseAttribute()
       #selects attribute that best separates Examples relative to Target
       let Tree = new decision tree with Best as root node

       foreach value v_i of Best:
              let Examples_i = subset of Examples that have Best=v_i
              let Subtree = ID3(Examples_i, Attributes-Best, Target)
              add branch from Tree to Subtree with label v_i

return Tree
```

Decision Tree example

- Example on behavioral disorder

Michielin, P., Pizzo, S., Cilia, S., Contino, C., Bettinardi, O., Bertolotti, G., Panzieri, A., Sanavio, E., Vidotto, G., Visentin, A. & Zotti, A. M. (2024). ***CBA 2.0 Primary Scales Battery: New Data and Interpretive Criteria for Depressive Disorders*** *(CBA 2.0 Scale Primarie: Nuovi dati e criteri interpretativi per le problematiche depressive)*. Psicoterapia Cognitiva e Comportamentale.

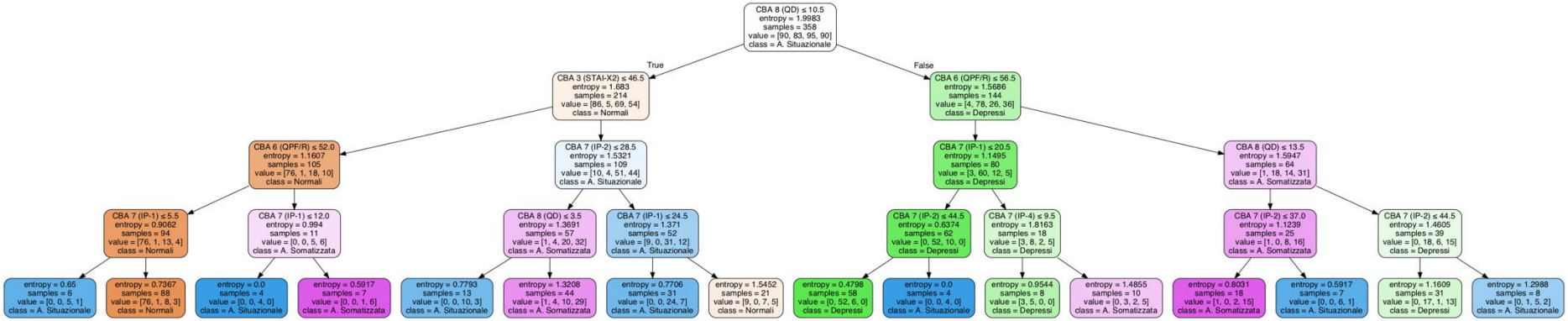
https://rivistedigitali.erickson.it/psicoterapia-cognitiva-comportamentale/en/archivio/vol-30-n-1/cba-2-0-scale-primarie/

Depression Detection

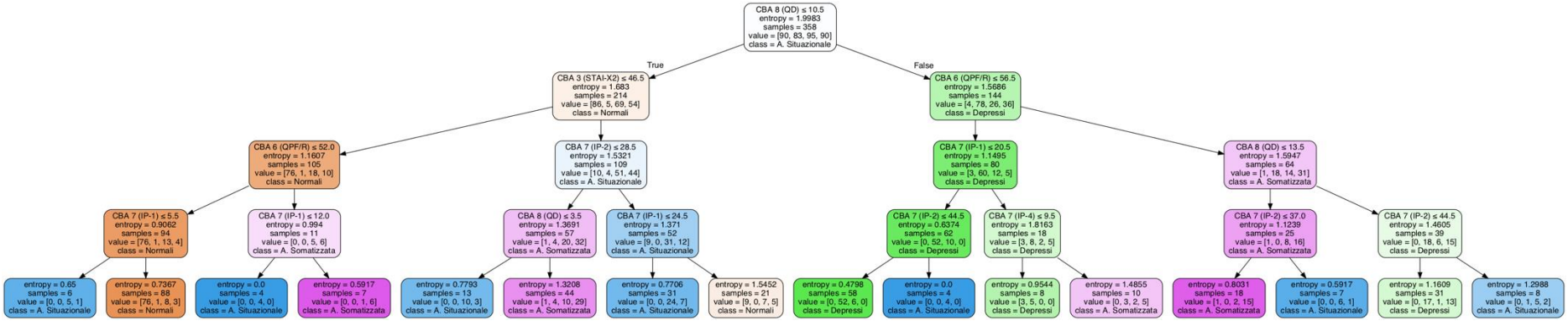
Inputs

- Age
- Gender
- CBA Scores
  - 22 different questionnaires each providing a score
- Reliability Score IR

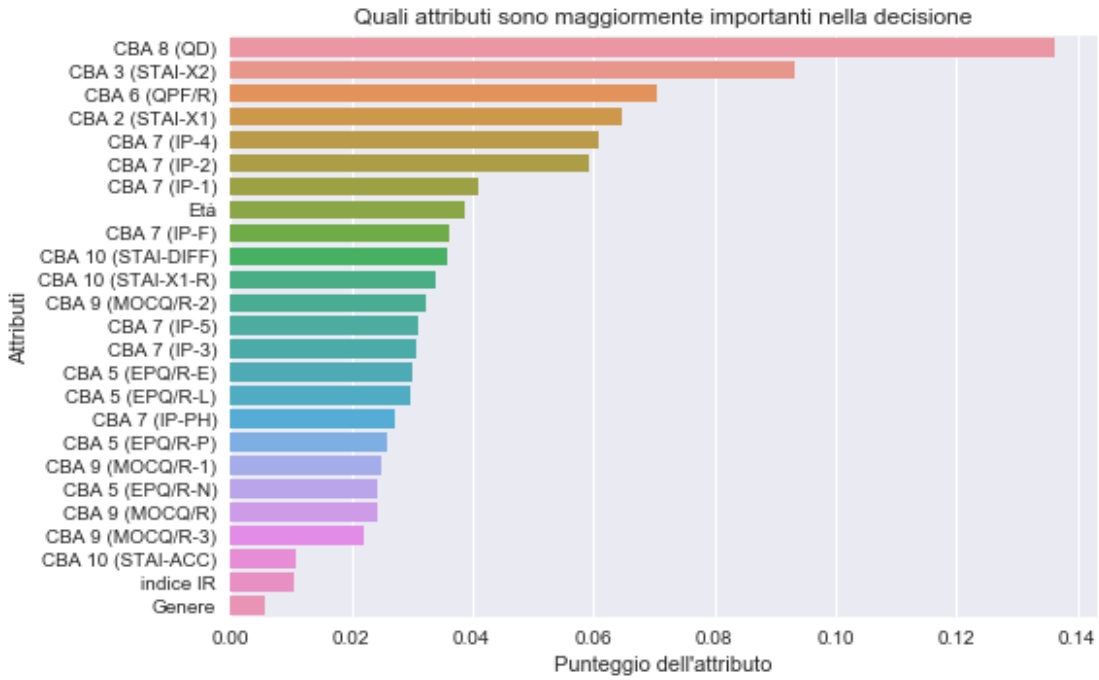
Depression Detection



Depression Detection



Depression Detection



That's all Folks!
SUPERB wallpapers